

MATH 55 FIRST MIDTERM EXAM, PROF. SRIVASTAVA
SEPTEMBER 29, 2022, 5:10PM–6:30PM, 2050 VLSB.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	12
2	8
3	5
4	8
5	8
6	9
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation.

(a) (3 points) The compound propositions **T F**

$$\forall x \exists y (P(x) \wedge Q(y))$$

and

$$\exists x \forall y (\neg P(x) \vee \neg Q(y))$$

are negations of each other.

Solution: True. By de Morgan:

$$\neg \forall x \exists y (P(x) \wedge Q(y)) \equiv \exists x \forall y \neg (P(x) \wedge Q(y)) \equiv \exists x \forall y (\neg P(x) \vee \neg Q(y)).$$

(b) (3 points) The compound proposition **T F**

$$(\exists y \forall x (P(x, y))) \rightarrow (\forall x \exists y (P(x, y)))$$

is a tautology.

Solution: True. Proof: Assume the first proposition. Choose y such that $\forall x (P(x, y))$. We will show the second proposition. Assume x is an arbitrary element of the domain. By our choice of y , $P(x, y)$ is true, so $\exists y P(x, y)$. Since x was arbitrary, the second proposition follows.

(c) (3 points) If $f : A \rightarrow B$ and S and T are subsets of A then **T F**

$$f(S - T) \supseteq f(S) - f(T).$$

(where $f(S) = \{b \in B : \exists s \in S \text{ s.t. } f(s) = b\}$ denotes the image of a subset of A).

Solution: True. Proof: Suppose $x \in f(S) - f(T)$. Then we can choose $s \in S$ such that $x = f(s)$ and $f(s) \notin f(T)$. By the definition of image, the latter statement implies $s \notin T$. Thus, $s \in S - T$ and $x \in f(S - T)$, so $f(S) - f(T) \subseteq f(S - T)$, as desired.

(d) (3 points) If $d, x, y, \in \mathbb{Z}_+$ and $d|xy$ then $d|x$ or $d|y$. **T F**

Solution: False. There are many counterexamples, e.g.: $x = 2 \cdot 3$ and $y = 3 \cdot 4$. Then $9|xy$ but $9 \nmid x, 9 \nmid y$.

[Scratch Space Below]

Name and SID: _____

2. (a) (4 points) **Prove or Disprove** the following statement: if $f : B \rightarrow C$ and $g : A \rightarrow B$ are functions and $f \circ g$ is one-to-one, then g must be one-to-one.

Solution: This statement is true. Proof: Assume $f : B \rightarrow C, g : A \rightarrow B$, and $f \circ g$ is one-to-one. Assume for contradiction that g is not 1-1. Choose $x, y \in A$ such that $x \neq y$ and $g(x) = g(y)$. Applying f , we find that $f(g(x)) = f(g(y))$, i.e., $f \circ g(x) = f \circ g(y)$. But this implies that $f \circ g$ is not 1-1, a contradiction.

I could also have written this proof using the contrapositive, which would have been essentially equivalent. One way you could have guessed that this proposition is true is by drawing a cartoon with three sets and arrows between them. The proof is saying that if there is a “collision” (two arrows with the same image) in $g : A \rightarrow B$ then there must be a collision in $f \circ g : A \rightarrow C$.

- (b) (4 points) **Prove or Disprove** the following statement: if $f : B \rightarrow C$ and $g : A \rightarrow B$ are functions and $f \circ g$ is one-to-one, then f must be one-to-one.

Solution: This statement is false. Counterexample: Let $A = C = \{1\}$ and $B = \{1, 2\}$ and define $g(1) = 1, f(1) = 1, f(2) = 1$. Then $f \circ g$ is 1-1 but f is not 1-1.

There are many possible counterexamples here. The idea is that elements in B outside $g(A)$ do not change whether or not $f \circ g$ is 1-1, but they do affect whether f is 1-1, so by adding some extra elements to B and appropriately defining f on $B - g(A)$, we can violate that f is 1-1.

Name and SID: _____

3. (5 points) Prove that if A is a countably infinite set and B is a finite set, then $A \times B$ is countably infinite.

Solution: Suppose $a_1, a_2, \dots, a_k, \dots$ is an enumeration of A and $B = \{b_1, \dots, b_n\}$. Consider the infinite array whose k th row for $k = 1, 2, \dots$ is

$$(a_k, b_1), (a_k, b_2), \dots, (a_k, b_n).$$

Notice that each element of $A \times B$ appears exactly once in this array: in particular, (a_k, b_j) appears as the j th element of the k th row. Since each row has finitely many elements, we can define the sequence

$$s = \text{first row of the array, second row of the array, } \dots$$

Since each element of $A \times B$ appears exactly once in this sequence, it is an enumeration of $A \times B$, so $A \times B$ is countably infinite.

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Name and SID: _____

4. (a) (4 points) Prove that if x and y are rational numbers such that $x < y$ then there exists a rational number z such that $x < z < y$.

Solution: Assume x, y are arbitrary rationals and $x < y$. Choose integers a, b, c, d such that $x = a/b$ and $y = c/d$, with $b, d \neq 0$. Let $z = \frac{x+y}{2}$ and observe that $z = a/b + c/d = (ad+bc)/bd$ is rational. Moreover, $z - x = \frac{x+y-2x}{2} = \frac{y-x}{2} > 0$ since $y > x$ so $x < z$. By a similar argument $z < y$. Thus, z is rational and $x < z < y$, as desired.

- (b) (4 points) Prove that if x and y are rational numbers such that $x < y$, then the set

$$S = \{z \in \mathbb{Q} : x < z < y\}$$

is infinite (here $\mathbb{Q}$ denotes the set of rational numbers. hint: use your solution to the previous part of the question.).

Solution: Let x, y be arbitrary rationals with $x < y$. Assume for contradiction that S is finite, and let

$$z_1 < z_2 < \dots < z_n$$

be all of its elements, sorted in increasing order. By the previous part of this problem, we can choose a rational z such that $z_1 < z < z_2$. In particular, $z \in S$ but $z \neq z_i$ for $i = 1, \dots, n$, contradicting our choice of $z_1, \dots, z_n$.

The idea of this proof is similar but not identical to the proof of the infinitude of primes: we assume for contradiction that there are finitely many, and then construct one more that is not in the list of all of them.

5. (a) (4 points) Calculate

$$(38)^{2022} \bmod 5.$$

Solution: *There are two different ways to do this : Fermat's little theorem and repeated squaring. I will do FLT since it is quicker.* Note that $38 \equiv 3 \pmod{5}$ (in particular, $38 \not\equiv 0 \pmod{5}$). By FLT we know that $38^4 \equiv 3^4 \equiv 1 \pmod{5}$. By the division algorithm, $2022 = 4 \cdot 505 + 2$, so by the rules of modular arithmetic

$$(38)^{2022} \equiv 3^{4 \cdot 505} 3^2 \equiv 1^{505} \cdot 9 \equiv 4 \pmod{5},$$

so the answer is 4.

- (b) (4 points) Prove that there do not exist integers
- $a, b, c \in \mathbb{Z}$
- such that

$$a^8 + b^8 + c^8 = 555554.$$

(hint: consider the remainder modulo 5 of both sides of this equation)

Solution: Assume for contradiction that there are integers a, b, c such that the above equation holds. Evaluating both sides modulo 5, we obtain the congruence

$$a^8 + b^8 + c^8 \equiv 555554 \pmod{5}.$$

Observe that either $a \equiv 0 \pmod{5}$ or by Fermat's little theorem, $a^4 \equiv 1 \pmod{5}$. In both cases, we have $a^8 \equiv 0 \pmod{5}$ or $a^8 \equiv 1 \pmod{5}$. The same argument implies that b^8 and c^8 must also be congruent to 0 or 1 modulo 5. Thus,

$$a^8 + b^8 + c^8 \equiv 0, 1, 2, \text{ or } 3 \pmod{5}.$$

However, $555554 \equiv 4 \pmod{5}$, so this congruence has no solution. Thus, no integers a, b, c , satisfying the original equation exist.

This problem is quite similar to a question on mt1prac1, but with a sum of three eighth powers instead of a sum of two squares. This is a general technique in number theory: to show that an equation has no integer solutions, you reduce it modulo a carefully chosen prime and show that the congruence has no solution, which is typically easier.

Name and SID: _____

[Scratch Paper 1]

Name and SID: _____

6. (a) (3 points) Use the Euclidean algorithm to calculate $\gcd(9, 44)$.

Solution:

$$44 = 4 \cdot 9 + 8$$

$$9 = 1 \cdot 8 + 1$$

So $\gcd(44, 9) = \gcd(9, 8) = \gcd(8, 1) = 1$.

- (b) (3 points) Use your answer to part (i) to find an integer $x \in \{0, 1, \dots, 44\}$ which satisfies

$$9x \equiv 5 \pmod{44}.$$

Solution: Rearranging the first two equations above, we have

$$1 = 9 - 1 \cdot 8 = 9 - (44 - 4 \cdot 9) = 5 \cdot 9 - 44.$$

Reducing modulo 44 we find that 5 is an inverse of 9 modulo 44, i.e., $9 \cdot 5 \equiv 45 \equiv 1 \pmod{44}$. Multiplying both sides of the congruence by 5, we have:

$$x \equiv 9 \cdot 5x \equiv 5 \cdot 5 \equiv 25 \pmod{44}.$$

Thus, $x = 25$ is the desired solution.

- (c) (3 points) Prove that there is no integer $x \in \mathbb{Z}$ which satisfies the congruence

$$8x \equiv 5 \pmod{44}.$$

Solution: Assume for contradiction that $8x \equiv 5 \pmod{44}$ for some integer x . Let x be such an integer. By the definition of congruence, this implies that $44 \mid (8x - 5)$. By the fundamental theorem of algebra, this means every prime factor of 44 is a prime factor of $8x - 5$. In particular, we must have $2 \mid (8x - 5)$. Observe also that $2 \mid 8x$. By the properties of divisibility, this implies that $2 \mid (8x - 5 - 8x)$, i.e., $2 \mid (-5)$, which is absurd.

Name and SID: _____

[Scratch Paper 2]

Name and SID: _____

[Scratch Paper 3]